

TOPIC NAME: _____

DAY: _____

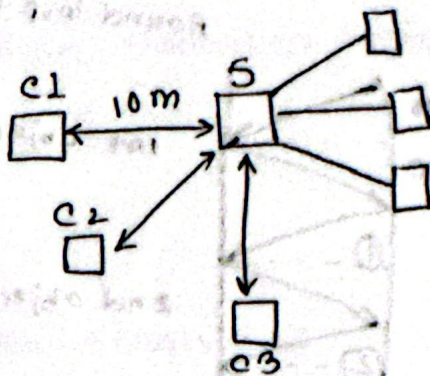
TIME: _____ DATE: / /

Application Layer:

Two methods:

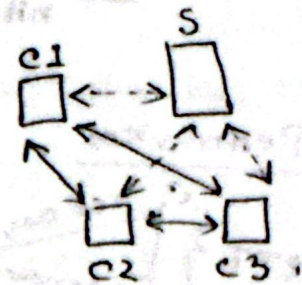
- i) client / server
- ii) peer to peer

client / server

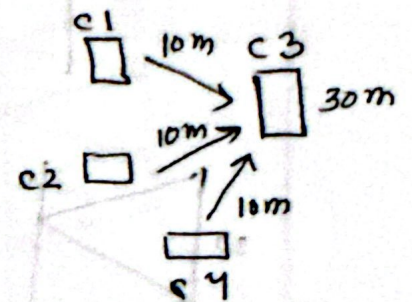


server mandatory
10m & max

peer to peer



server optional



2nd peer (not 2nd 3rd)

Web Page

http : // www . nytimes . com / tech / index . html

application host domain top path file
transfer name level
protocol domain

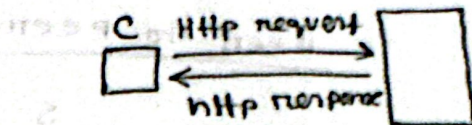
TOPIC NAME : _____

DAY : _____

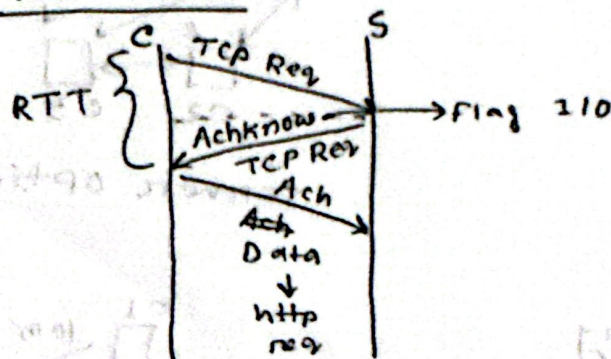
TIME : _____ DATE : / /

http → security नाग ना

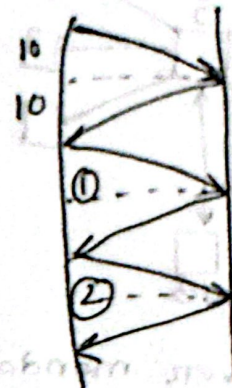
https → login करता. वर- there are private information,
उनका https करी. करी,



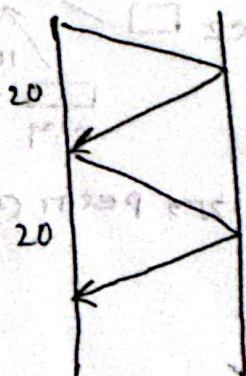
Persistent



Round trip time



$$\begin{aligned}
 \text{1st object} &= 2 \times \text{RTT} \\
 &= 2 \times 20 \\
 &= 40 \\
 \text{2nd object} &= 40 + 20 \\
 &= 60
 \end{aligned}$$



$$\text{obj} = 5 \text{ ms}$$

obj download time

$$= 20 + 20 + 5$$

GOOD LUCK

TOPIC NAME: _____

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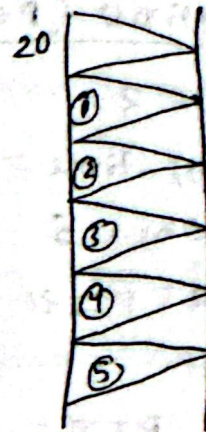
* $obj = 5$

$obj\ time = 5\ ms$

$RTT = (RTT \times 5) + RTT$

$= (1 + obj) RTT$

$= (1 + 5) 20 = 120\ ms$



File transfer time = Total RTT + Total obj time

$= 120 + (5 \times 5)$

$= 120 + 25 = 145\ ms$

Non-persistent

$obj\ -t = 5\ ms$

$obj = 5$

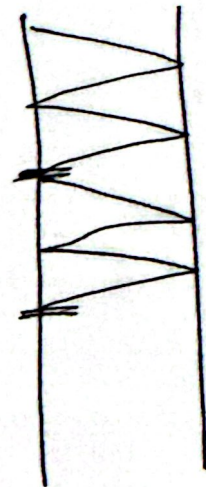
$RTT = 20\ ms$

$RTT = obj \times 2 \times RTT$

$= (5 \times 2) \times 20 = 10 \times 20$

$= 200$

$FTT = 200 + 25 = 225$



TOPIC NAME: _____

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Pipeline (Persistent)

$$P = 2$$

$$\text{obj-time} = 5 \text{ ms}$$

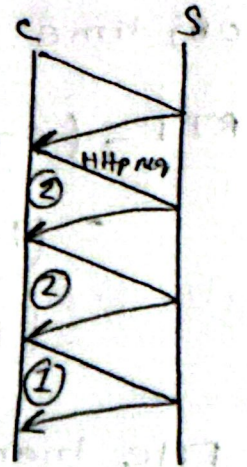
$$\text{obj} = 5$$

$$\text{RTT} = 20 \text{ ms}$$

$$\text{Total RTT} = 4 \times \text{RTT} = 80$$

$$\text{FTT} = 5 + 5 + 5 + 80 = 95$$

$$\text{obj}_e = \left\lceil \frac{\text{obj}}{e} \right\rceil = \left\lceil \frac{5}{2} \right\rceil = \lceil 2.5 \rceil = 3$$



TOPIC NAME: Non-persistent

DAY: _____

TIME: _____ DATE: / /

* RTT = 30 ms, Obj = 6, each obj = 15 MB, speed = 1.5 Mb/s

$$\begin{aligned}\text{Total RTT} &= \text{TCP RTT} \times 6 + \text{HTTP RTT} \times 6 \\ &= 30 \times 6 + 30 \times 6\end{aligned}$$

$$= 360 \text{ ms}$$

$$\begin{aligned}\text{Total transmission time} &= \frac{6 \times 15 \times 8}{1.5} = 480 \text{ ms} \\ &\quad \begin{array}{l} \text{total size} \\ \text{speed} \end{array}\end{aligned}$$

$$\begin{aligned}\text{Total response time} &= 360 \text{ ms} + (480 \times 1000) \text{ ms} \\ &= 480360 \text{ ms}\end{aligned}$$

Persistent HTTP

* Total object = 15, size = 10 MB. client download speed = 8 Mb
server upload speed = 10 Mb. TCP C-S = 15 ms,
HTTP S-C = 20 ms

$$\begin{aligned}\text{Total RTT} &= \text{TCP RTT} \times 1 + \text{HTTP RTT} \times 15 \\ &= 15 \times 2 \times 1 + 20 \times 2 \times 15 = 630 \text{ ms}\end{aligned}$$

$$\text{Total file transmission time} = \frac{10 \times 15 \times 8}{8} = 150 \text{ ms}$$

↳ lowest use server speed
client speed < server speed

$$\text{Total response time} = 630 + 150 = 780 \text{ ms}$$

$$2m \text{ 02T} = 0.81 + 0.25 = 1.06 = 1.06 \text{ sec}$$

$$\text{Total time} = \frac{8 \times 21 \times 0.1}{8} = 2.1 \text{ sec}$$

$$= 12 \times 5 \times 1 + 30 \times 5 \times 12 = 1.80 \text{ sec}$$

Fetch the object $\rightarrow$ T include $\rightarrow$ T

Webpage load time (after fetching IP) = T_{CP} + T_{TP} + T_T

Webpage load time on total response time

$$= \text{DNS} + \text{TCP} + \text{HTTP} + \text{T}$$

Transition time = Obj x download time

client response time

HTTP response = Database time + server to

Fetch request to DNS to request obj

DNS time (not request to server)

after fetching IP address $\rightarrow$ T

$$= 30 \times 5 + 30 \times 5$$

delay 5 T (server side)

delay 4 T (client side)

delay 5 T (server side) - 2m 02T = T_{CP} + T_{TP} + T_T

TOPIC NAME :

DAY :

DATE :

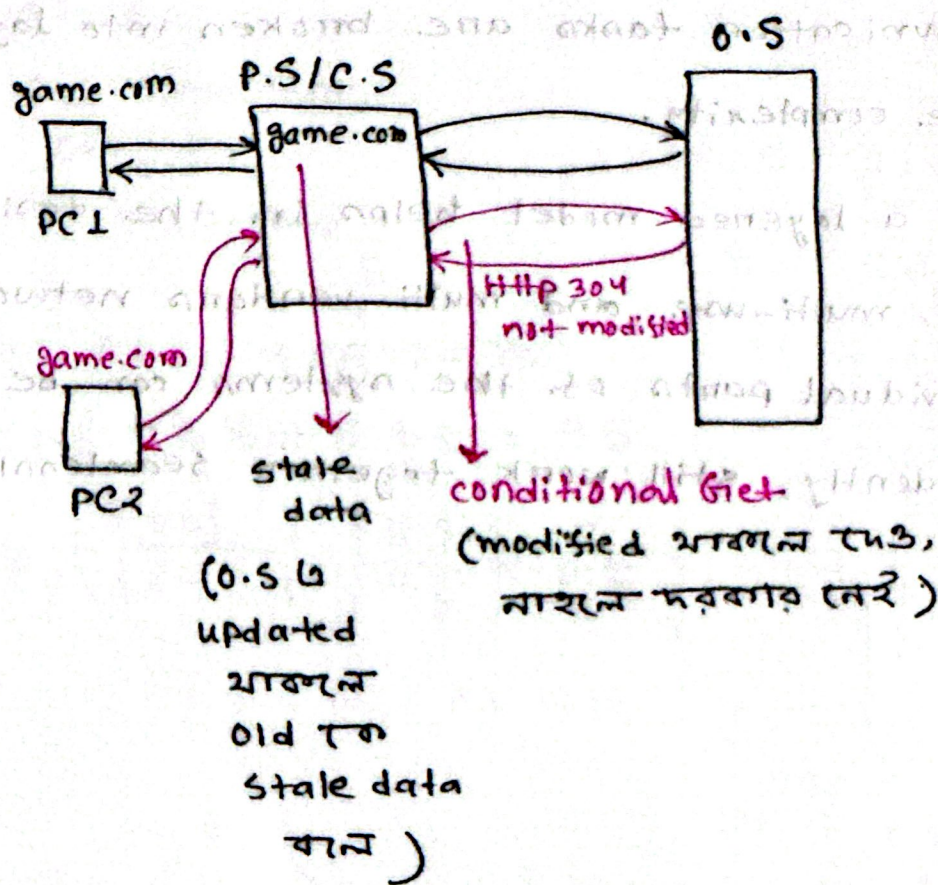
TOPIC NAME :

DAY : 9

TIME :

DATE : / /

Web Cache or Proxy Servers



proxy server এর function :

- track করতে পারে
- speed control করতে পারে
- proxy server act as a filter as well.

GOOD LUCK

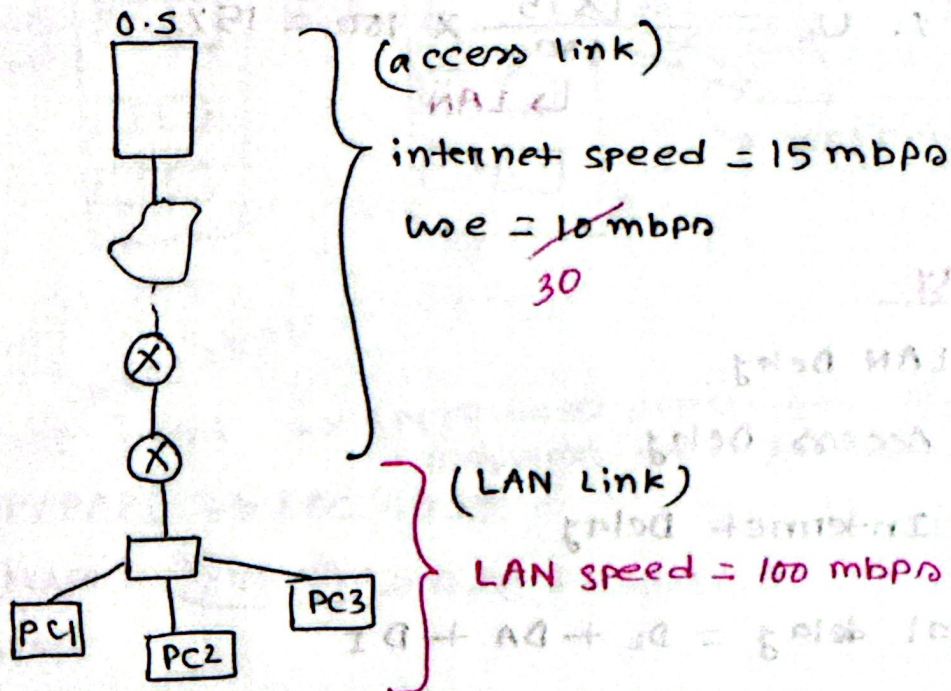
TOPIC NAME: _____

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Access Utilization



$$\text{utilization} = \frac{\text{use}}{\text{total}} \times 100 = \frac{10}{15} \times 100 = 67\%$$

$$\text{obj-size} = 1 \text{ M}$$

$$\text{obj-rate} = 15 \text{ s}^{-1}$$

$$\% \text{ UA} = \frac{\text{used bandwidth}}{\text{total bandwidth}} \times 100$$

$$= \frac{\text{obj-size} \times \text{obj-rate}}{\text{total bandwidth}} \times 100 = \frac{1 \times 15}{15 \times 30} \times 100$$

$$= 100\% \cdot 50\%$$

Speed वास्तव में costly.

TOPIC NAME: _____

DAY: _____

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After implementing proxy server:

$$1. U_L = \frac{1 \times 15}{100} \times 100 = 15\%$$

→ LAN

Delay

i) LAN Delay

ii) Access Delay

iii) Internet Delay

$$\text{Total delay} = D_L + D_A + D_I$$

$$* D_L = 5 \text{ ms}, D_A = 25 \text{ ms}, D_I = 3 \text{ s} = 3000 \text{ ms}$$

LAN Hit rate = 30%.

$$\text{total delay} = D_L + (D_A + D_I)$$

$$= 0.3(5) + 0.7(25 + 3000)$$

$$= 2119 \text{ ms}$$

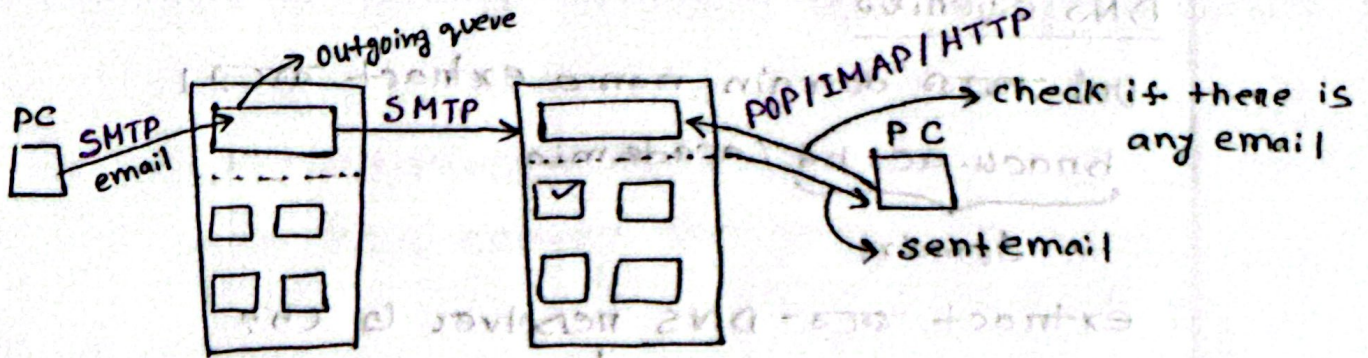
$$= 2.119 \text{ sec}$$

GOODLUCK

TOPIC NAME: Application Layer (Electronic Mail & DNS)

DAY: _____

TIME: _____ DATE: / /



Protocol

- SMTP } sent → RFC 2821
- POP/POP3 } receive → RFC 1939
- IMAP } RFC 1730
- HTTP }

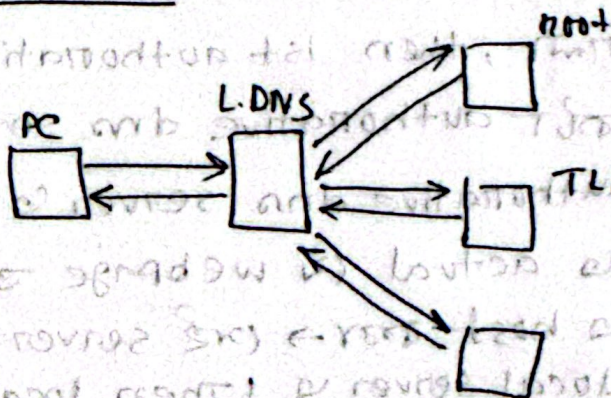
HTTP : Pull

↳ server থেকে client এ আর request করার পর data আসা শুরু করলে

SMTP: push

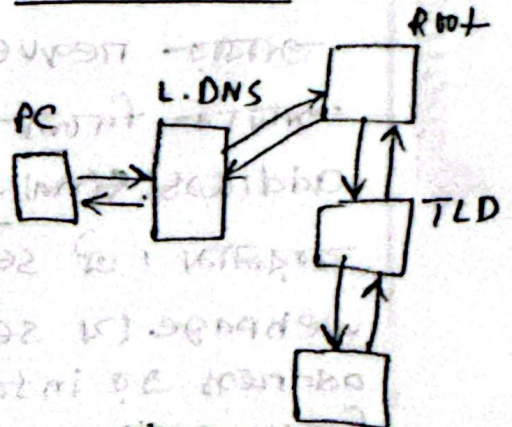
↳ request করার পর data যাওয়া শুরু করলে

Iterative



৭টি RTT

Recursive



৭টি RTT

TOPIC NAME: _____

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DNS Queries

URL থেকে domain name extract করে।

bnacu.ac.bd / academia

domain

extract করে DNS resolver ও হয়



operating system এর মধ্যে থাকে

Resolver check করে bnacu.ac.bd এর IP address
store করা আছে নাকি।

যদি cache এর মধ্যে IP address save করা না থাকে,
তখন resolver request করে local DNS server এর
কাছে।

Local DNS server তার নিজের cache তে check
করে এই particular web page এর কোনো information
আছে নাকি। Local DNS server top level domain
(.bd) এর information রাখতে পারে। যদি তার
কাছেও না থাকে তখন root name server এ যাবে।

www.net.compsci.googleplex.edu
edu তে জানা, edu 1st authoritative server এর info
দিয়বে। IP address পাওয়ার পর local dns server

তার request করানো, then 1st authoritative server
আবার দিলে আরেকটি authoritative dns server এর
address. Finally 3rd authoritative dns server এ request
করানো। এ server এ actual (যে webpage আছে, যে
webpage এ server এ host করে → সেই server এর IP
address এর info মার local server এ। Then local server
নিজের cache update করে (edu এর info save করে রাখবে।)

TOPIC NAME: _____

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Then DNS resolver এ পাঠানো, then resolver আমার PC কে পাঠাবে।

Process → Iterative query : কারণে

কারণ Local DNS server এর query করতে।

DNS: caching:

এ response আসে, server মটকে cache করতে এবং resolver ও cache করে। নির্দিষ্ট সময়ের জন্য cache to store করে। কারণ একটি webpage এ 1 host জানা, তারপর আর কোনো মাইনি, আই-আইসিআই-এর জন্য store করার প্রয়োজন নেই- এবং IP address এ 3 change হতে পারে।

TTL: নির্দিষ্ট সময়ের জন্য device এ store করে রাখি।

Local caching: Top-level domain এর record store করে রাখি।

Out of Date cache: Is a domain's IP address changes

Updating Standard: into their IP address changes।

TOPIC NAME : _____

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DNS Records

Resource records (RR)

RR format: (name, value, type, ttl)

Same host name same server to represent

→ CNAME

webpage & mail server → type MX

value → IP address = type A

value → dns server info = type NS

value → webpage & other url = type CNAME

value → mail = type MX

Electronic Mail

Three major components:

i) user agents → email read or write

ii) mail server → email store

iii) SMTP → email transfer & protocol

TOPIC NAME: _____ DAY: _____

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IP1, TTL = 5 hours

1st time 21st June 2023 9:18:00 PM

2nd time 23rd June

↓

expired → dns hierarchy reply fir

if iterative → DNS RTT = 4

non expired → DNS RTT = 1

fetch IP means DNS RTT = 4

from Idm → DNS RTT = 3

1 tr RTT = PC → Idm

3 tr RTT = Idm → dns hierarchy

→ Idm reply fir

PC → Idm → dns hierarchy (root, fld, as)

root → fld → as

iterative

PC → Idm → root → Idm → fld → Idm → as → Idm → PC

advantage:

i) idle bhar shakte hai na

ii) upper level ka burden create karta na

disadvantage: security issue

recursive

PC → Idm → root → fld → as → fld → root → Idm → PC

TOPIC NAME: Introduction to Transport

DAY: _____

Layer

TIME: _____

DATE: / /

Transport layer: Process to process delivery

যাতে device transport layer to access করে না।

Port address $\rightarrow$ 16 bit

Three categories:

i) well-known (0-1023)

ii) Registered (1024-49151)

iii) Dynamic / private (49152-65535)

Well-known: World wide use হয়।

Organization fix করে হয়।

Registered: A person can register the port address.

Register করলে overlap হয় না।

Dynamic / private: We usually use this.

GOOD LUCK

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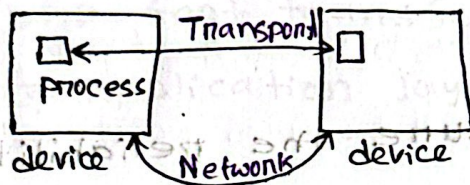
Two Transport Layer Protocol:

i) UDP : User Datagram Protocol

ii) TCP : Transmission Control Protocol

Transport layer : Logical communication between processes

Network layer : logical communication between hosts.



Functions of the Transport Layer

Primary responsibilities:

1. Segmentation: কোনো data lost হয়ে- তোলে খুঁজু সেই data পাঠাবে, পুরাতা পাঠাবে হতে না। and managing each piece.

২. Reassemble: যেভাবে data পাঠিয়েছে, সেভাবে যেন receive করতে পারি।

3. Identifying the different application
identify করতে port address use করতে হয়।

৪. Multiplexing: different processes network layer কে at the same time access করতে পারবে OR data পাঠাতে পারবে।

TOPIC NAME : _____

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5. Initiating and terminating a session:



call કરે એક બીજા

then message

જાહેર -

6. flow control → receiver સુધી નિયંત્રિત જાય છે, (અહીં)

- બહુ ઝાડા data જાય છે,

7. Error recovery



This functions ensure the reliability of data transfer.

Reliable → TCP (Ordered)

Fast → UDP

UDP

Best effort service

Used by applications that requires no delay in data delivery.

How does UDP deliver fast?

i) no connection establishment

ii) small header size

iii) no error or flow or congestion control; UDP can blast away as fast as desired.

GOOD LUCK

TOPIC NAME: _____

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UDP is used by:

i) streaming multimedia apps

ii) SNMP

iii) DNS

iv) HTTP/3

→ TCP, UDP দুইটিই we করে

} used sometimes

Reliable transfer over UDP:

এক্ষেত্রে- UDP সঠিক transfer করে, আর- reliability handle করে application layer এ, যা http provide করে।

Segmentation: যে data আছে, সেটাকে segment করা লাগে।

TCP on UDP সঠিক করে করা লাগবে। UDP হলে UDP এর header use করবে, TCP হলে TCP এর header use করবে।

Reassembly: sequence number দিয়ে reassemble করি।

UDP এর ক্ষেত্রে reassemble করে না।

TCP reassemble করে application layer এ পাঠায়।

Reassembly ২ ভাগে ^{হয়} ~~হয়~~ হয়। UDP order maintain করে না, TCP order maintain করে।

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TCP and UDP headers

UDP header is sequence number, and reassembled order is not required.

UDP is total length = header + data
↳ payload

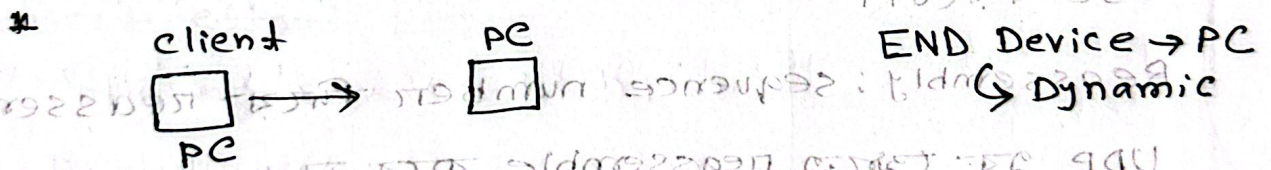
Identifying Different Application

Identify port number first.

↳ 16 bit → 0-65535

Client side is well known port address use
if not, dynamic port address use, and
randomly generate.

Server side is http is 80, DNS is 53
port number use.



Request

Source destination
Server Dynamic registered

Destination PC is registered port address use
required.

TOPIC NAME: _____ DAY: _____
TIME: _____ DATE: / /

Request আমর uniquely identify করি- socket address
(IP address: Port) দ্বারা।

Multiplexing: Different application থেকে data নিয়ে network layer এ পাঠিয়ে দেয়া হয়। Transport layer বিভিন্ন application থেকে data নিয়ে পাঠিয়ে দেয়।

Demultiplexing: port address বা socket address দ্বারা demultiplexing করা হয়। Demultiplexing এর জন্য at the same time multiple application use করা হয়।

Transport layer network layer কে directly কিছু পাঠায় না, socket এর মাধ্যমে পাঠায়। Demultiplexing এর ক্ষেত্রে socket use করা হয়।

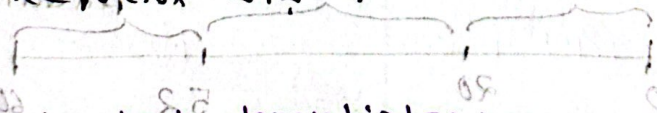
Connectionless demultiplexing: → UDP

UDP র socket কে দুটো চিহ্নিত দ্বারা identify করি।

i) destination IP address (ii) destination port address

connectionless demultiplexing এ একটা socket থেকে,

প্রাপ্ত সব request অন্য-



Connection-oriented demultiplexing:

Tcp socket identified by 4 tuple:

i) source IP address, ii) source port number

iii) dest IP address, (iv) dest port number

TOPIC NAME: Transport Layer (TCP)

DAY: _____

TIME: _____

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TCP Segment Header

Sequence Number: যখন unordered data segment to reassemble করতে, তখন লাগবে।

TCP header = 60 bytes

UDP header = 8 bytes (network overhead নেই, no need to ensure reliability).

TCP header

→ 60 bytes

20 bytes always fixed

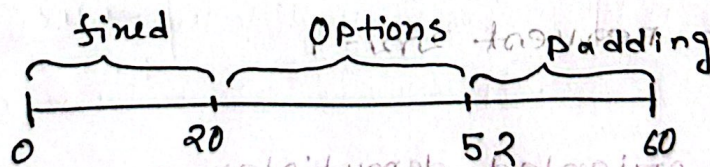
40 bytes → Options
padding

HLEN → (0-15)

$$* HLEN = (1101)_2 = (13)_{10} = (13 \times 4) = 52 \text{ bytes}$$

$$\text{Options} = 52 - 20 = 32 \text{ bytes}$$

$$\text{Padding} = 60 - 52 = 8 \text{ bytes}$$



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Transport layer દ્વારા આ object application layer ૫ ઇલે

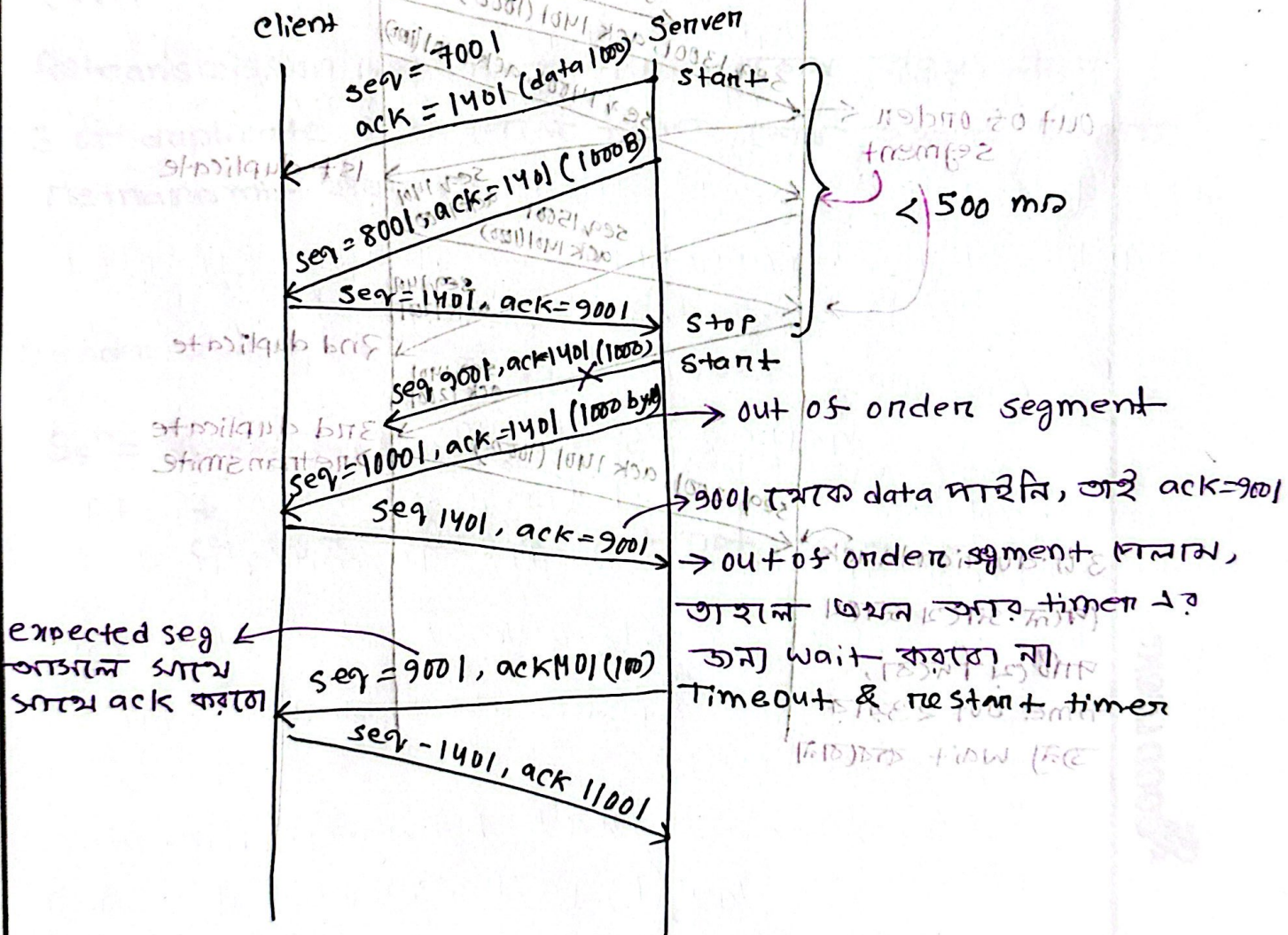
ટોટલ window size reset રહે- રહ્યું.

Window size કમીટી થાતો, યથા data consume કરે.

Urgent pointer

sequence number + urgent pointer value = last urgent byte

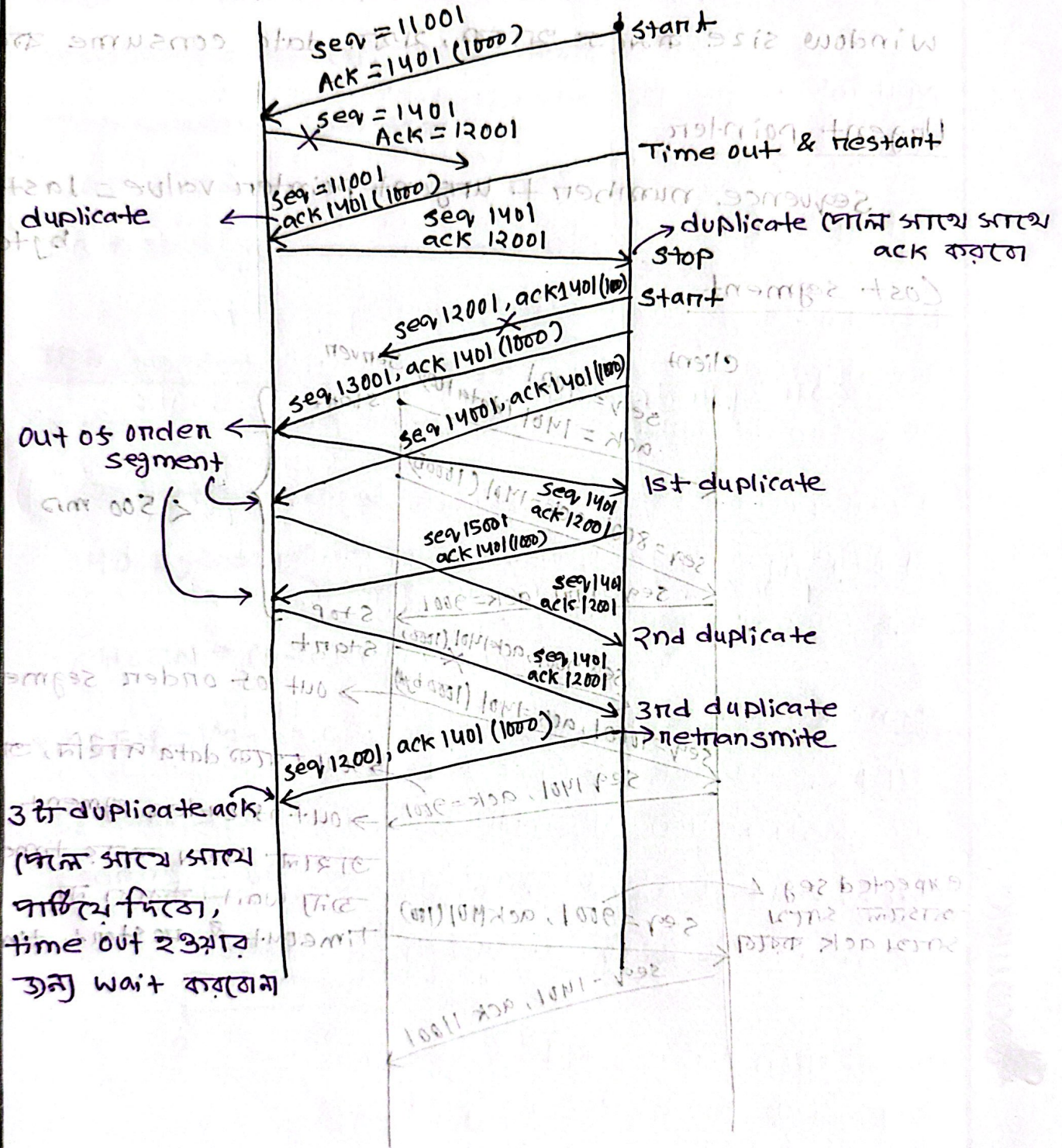
Lost segment



DAY: _____

TIME: DATE: / /

Lost Ack



TOPIC NAME: _____

DAY: _____

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DATE: / /

Acknowledgement পাঠাই timer কোথায় স্থান

exception

i) multiple in order segment

ii) out of order segment

iii) duplicate segment

iv) missing segment

সময় সময়
~~যখন time শেষ হওয়া যায়~~ ack করতে।

Retransmission পর সময় সময় কোথায় হওয়া লাগে।

3 টি duplicate ack যখন সময় সময় কোথায় হওয়া যায়
retransmit করতে।

Sender Sliding Window

S_s = first outstanding byte

↓
যে byte পাঠিয়েছি, but not acknowledged

9:30

TOPIC NAME : _____

DAY : _____

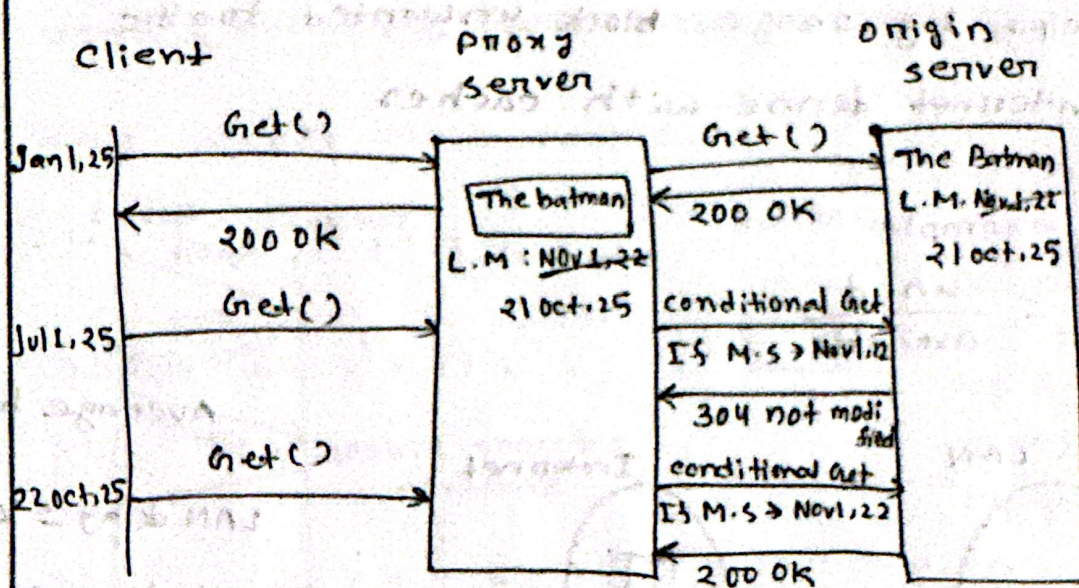
TIME : _____

DATE: 21/ 10 / 25

Stale Cache

conditional get

*



HTTPS → secured

Public key → encrypt કરી, user એ માટે થાય

private key → server એ થાય, સૂચક સંચાર છે decrypt કરવા થાય.

Encryption protocol → Transport Layer Security

GOOD LUCK™

Imp for Mid

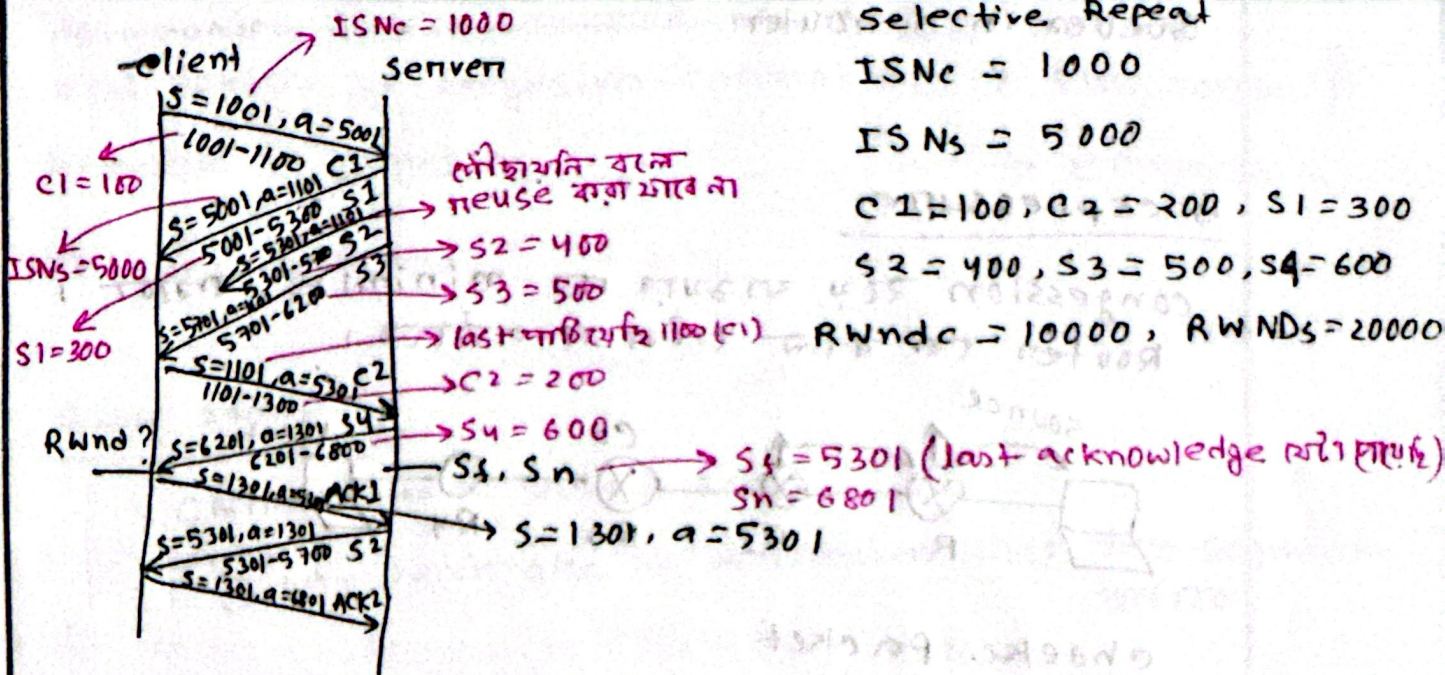
9:30

TOPIC NAME: TCP

DAY: _____

TIME: _____

DATE: 5 / 11 / 25



Rwnd = 10000

- 300

- 500

- 600

8600

S:

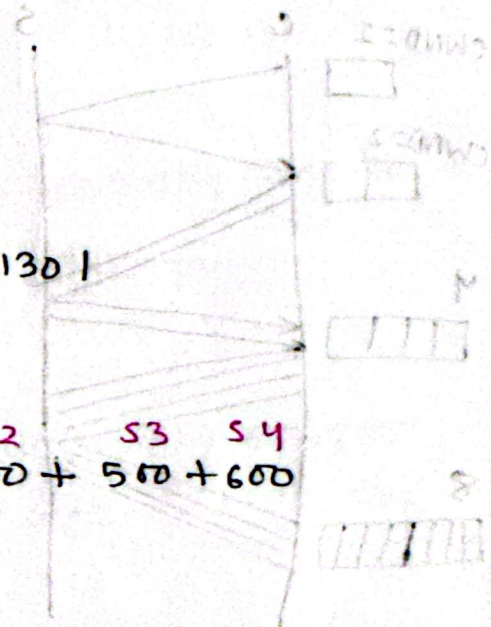
$$1000 + 1 + 100 + 200 + 0 = 1301$$

ISNc SYN C1 C2 ack

Ack:

$$\begin{aligned} \text{ack} &= \text{ISNs} + \text{SYN} + \text{S1} + \text{S2} + \text{S3} + \text{S4} \\ &= 5000 + 1 + 300 + 400 + 500 + 600 \\ &= 6801 \end{aligned}$$

pointed moment এ S2 পাঠানি, তাই minus করতে না।



GOOD LUCK